KALPA: OMNICHANNEL UNIT ECONOMICS AND MEMBERSHIP LIFETIME VALUE IN HYBRID RETAIL

> *"In legacy retailing, growth was often bought through discounting and geographic expansion. In a hybrid omnichannel and membership ecosystem, growth without diagnostic unit-level discipline is merely an acceleration toward margin collapse."*

Abstract

This case study examines the strategic, operational, and analytical transformation of KALPA, a hybrid retail and subscription enterprise operating physical outlets alongside digital touchpoints (Mobile App and Web). Confronting margin erosion across its multi-channel commerce engine and acute churn within its tiered recurring membership program, KALPA's leadership sought to transition from siloed channel reporting to an integrated data-driven operating model. Drawing inspiration from Arvind Fashions' pioneering omnichannel playbook and its "One Customer, One Inventory" architecture, this study evaluates the mechanics of KALPA's customer value creation, unit economics waterfall, and subscription renewal funnels.

Through an audit of KALPA's nine-table relational architecture, the narrative resolves critical analytical pitfalls—including top-line distortions from uncompleted orders, right-censoring in subscription renewal horizons, and sample skews introduced by institutional B2B procurement. By mapping findings against classical management frameworks—including Porter's Five Forces, activity-based contribution modeling, and multi-dimensional cohort survival—the case details how KALPA untangled voluntary customer disengagement from technical payment failure, quantified the non-business omnichannel customer multiplier, and established a phased roadmap for profitable digital-first retail orchestration.

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1. The Dilemma at KALPA

The morning sun filtered through the conference room blinds at KALPA's headquarters, illuminating a dense array of Looker Studio dashboards and financial spreadsheets. The executive leadership team sat in focused silence, reviewing the latest performance figures across the company's operating channels—physical stores (**STORE**), mobile commerce (**APP**), and digital storefronts (**WEB**)—alongside its recurring subscription membership program (**plan_code**).

On the surface, the enterprise projected vibrant multi-channel growth. Order volumes across the digital properties were climbing, and the membership roster indicated thousands of enrolled subscribers. However, beneath the aggregate revenue figures lay uncomfortable operational realities:

1. **Unit Economics and Profitability Compression:** Despite healthy top-line gross receipts, contribution margins after accounting for markdown discounting, merchandise cost of goods sold (COGS), order fulfilment, and payment gateway levies were tightening month-over-month. Top-line expansion was not translating into commensurate net cash flow.
2. **Channel Role Ambiguity:** Internal debates persisted regarding capital allocation. Was the mobile app cannibalizing the physical store base, or was it cultivating a high-value customer profile? More critically, did the multi-channel shopper justify aggressive customer acquisition expenditure, or were reported spend premiums an artifact of bulk procurement from the commercial business segment?
3. **Membership Churn and Subscription Leakage:** The recurring renewal funnel appeared to be deteriorating. Initial management reviews of the 2026 renewal opportunities indicated a conversion rate hovering around 50%. It remained unresolved how much of this churn was attributable to genuine consumer dissatisfaction versus technical gateway friction, mandate expirations, or flawed analytical reporting.

The managing leadership reflected on the strategic playbook of **Arvind Fashions Limited (AFL)** (IIMA/IS0147). Under the stewardship of Executive Director Kulin Lalbhai and Chief Digital Officer Nitesh Kanchan, Arvind had confronted severe marketplace disintermediation, margin dilution from aggressive third-party e-commerce discounting, and an absence of end-consumer visibility. Arvind had engineered an operational turnaround by implementing a unified backend ("One Customer, One Inventory"), deploying algorithmic pricing guardrails, and demonstrating that an omnichannel customer delivered a **2.5x spend multiple** over single-channel shoppers while driving down blended Customer Acquisition Cost (CAC).

Could KALPA replicate Arvind's structural discipline within a hybrid business model that married traditional retail transactions with subscription memberships?

To address this question, KALPA's data science and commercial operations teams initiated a granular audit across the company's enterprise data lake, evaluating transactional logs, cost accounting structures, and membership interaction events.

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2. Business Context and Organizational Model

Unlike Arvind Fashions—a multi-brand apparel manufacturing and retail franchise conglomerate managing over 1,300 Exclusive Brand Outlets (EBOs), 5,000 Multi-Brand Outlets (MBOs), and third-party marketplace storefronts (Myntra, Flipkart, Ajio, Amazon)—KALPA operates a hybrid business model. It unifies proprietary direct-to-consumer commerce with an ongoing subscription program across three direct channels:

- **STORE:** The brick-and-mortar retail network providing physical touchpoints, immediate product handovers, and in-person customer service.

- **APP:** The proprietary native mobile application designed to drive repeat engagement, real-time push communications, and friction-free mobile transacting.
- **WEB:** The browser-based direct-to-consumer digital portal serving desktop and mobile web shoppers.

Complementing this retail triad is KALPA's tiered **Membership Program**. Members subscribe under designated tiers (`plan_code`) sourced through distinct promotional funnels (`acquisition_channel`). Membership grants consumers ongoing transactional privileges and curated benefits, requiring annual renewal cycles (`renewal_opportunities`) managed via systemic communication sequences and payment capture protocols (`member_events`).

KALPA's commercial footprint spans two primary customer categories captured in the customer master:

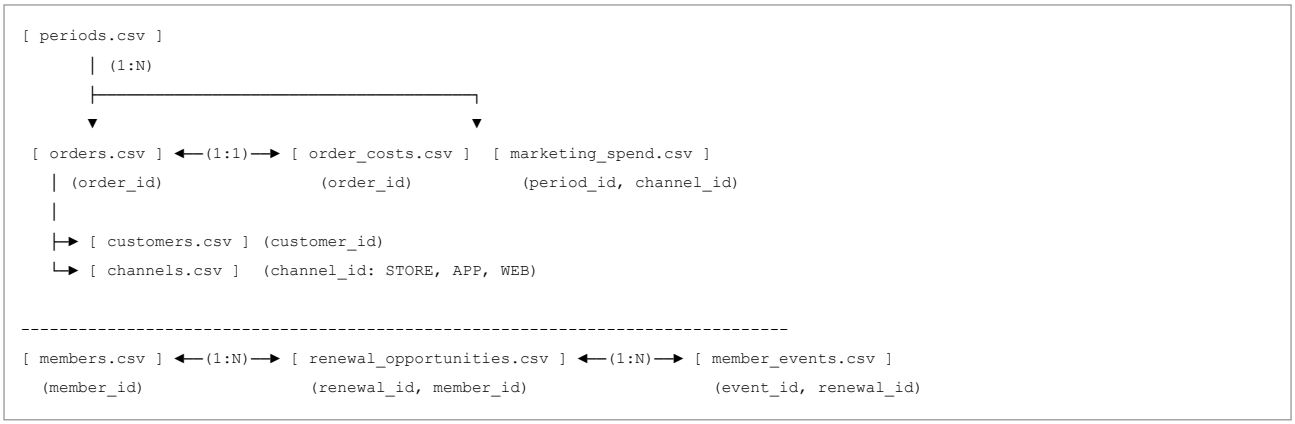
1. **Consumer Segments:** Everyday retail and direct consumers purchasing personal items across physical and digital formats.
2. **Business Segment (`segment == "Business"`):** Institutional and B2B corporate buyers executing bulk purchasing agreements, commercial orders, and enterprise-scale transactions.

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3. The Analytical Architecture and Data Integrity Foundations

To evaluate performance, KALPA relies on a relational architecture spanning nine primary data tables. Before any managerial conclusions could be drawn, the analytics team had to audit the integrity of the pipelines, isolating data quality pitfalls and logic traps that had previously led leadership to faulty strategic conclusions.

3.1 Relational Schema



1. `orders.csv`: Granular order header capturing `order_id` (PK), `customer_id` (FK), `channel_id` (FK), `period_id` (FK), `order_date`, transactional `status` (including `COMPLETED`, `CANCELLED`, and ancillary states), `gross_amount`, `discount_amount`, and `refund_amount`.
2. `order_costs.csv`: Financial unit cost ledger joined 1:1 on `order_id` (PK/FK), tracking `product_cost` (COGS), `fulfilment_cost` (outbound logistics and pick-pack), and `payment_cost` (gateway commissions and settlement fees).
3. `customers.csv`: Customer registry housing `customer_id` (PK) and demographic categorization `segment` (differentiating standard retail consumers from institutional "Business" accounts).
4. `channels.csv`: Touchpoint registry containing `channel_id` (PK), cataloging `STORE` (offline) alongside `APP` and `WEB` (collectively tracked as `DIGITAL`).
5. `marketing_spend.csv`: Commercial advertising ledger detailing expenditures by `period_id` (FK), `channel_id` (FK), operational `purpose` (Acquisition vs. Retention/Brand), and monetary `amount`.
6. `periods.csv`: Temporal master cataloging `period_id` (PK) and macro calendar definitions.
7. `members.csv`: Master subscriber registry detailing `member_id` (PK), historical membership status at baseline (`status_2025_10_09`), ongoing status (`status_2026_10_09`), assigned subscription tier (`plan_code`), and originating source (`acquisition_channel`).
8. `renewal_opportunities.csv`: Contractual renewal milestone records tracking `renewal_id` (PK), linked `member_id` (FK), historical baseline `expiry_date`, final operational `renewal_deadline`, and sequential lifecycle ordinal `renewal_number` (distinguishing first renewals from subsequent tenures).
9. `member_events.csv`: Granular event log registering operational renewal actions: `event_id` (PK), linked `renewal_id` (FK), chronological `event_time`, functional `event_type` (`PAYMENT_ATTEMPT`, `REMINDER_SENT`), gateway/delivery `outcome` (`SUCCESS`, `FAILED`), machine `detail_code` (diagnostic error strings), and operational `event_cost`.

3.2 Methodological Adjustments & Data Integrity Corrections

Prior executive dashboards had reported contradictory metrics, leading to organizational friction between finance, channel heads, and marketing. The data audit isolated three critical methodology corrections:

1. The Order Status Distortion (Completed vs. Gross Volume)

- **The Trap:** Previous commercial summaries incorporated non-completed and cancelled order records into top-line turnover reporting.
- **The Diagnostic:** Incomplete orders inflated headline gross volume while failing to account for downstream cancellations. Consequently, cross-functional P&L statements disagreed across divisional views.
- **The Correction:** All commercial, channel, and omnichannel P&L algorithms strictly filter for realized economic transactions: $\text{Target Population} \iff \text{status} == \text{"COMPLETED"}$ Cancelled orders are removed from revenue waterfalls and isolated as an operational drag indicator.

2. The Right-Censoring Trap in Membership Renewals

- **The Trap:** When assessing the 2026 renewal cohorts, naive reporting evaluated successful conversions as the simple share of renewals logging an accepted payment attempt against total observed renewal records. This pegged the first-renewal conversion rate at approximately **50%**, triggering alarm over cohort retention collapse.
- **The Diagnostic:** The data snapshot had a fixed analytical boundary: $\text{as_of} = \max(\text{member_events.event_time})$ A substantial volume of active subscribers possessed a contractual renewal deadline falling *after* the analytical cutoff date: $\text{is_open} \iff \text{renewal_deadline} > \text{as_of}$ These members were not lost; they were still within their active renewal window. Treating open renewals as failed expirations depressed cohort performance.
- **The Correction:** Filtering strictly to completed renewal horizons (`~ren["is_open"]`): $\text{Closed Renewal Conversion Rate} = \frac{\sum \text{is_success}}{\text{Count}(\text{Closed Renewals})}$ This adjustment established that KALPA's true 2026 first-renewal rate stood at **~55%**—five percentage points higher than naive assessments indicated.

3. Entity Siloing and Institutional Segment Skew

- **The Trap:** Initial analytics attempted to evaluate omnichannel customer value across the raw customer registry while treating subscription renewals as an extension of retail order behavior.
- **The Diagnostic:** Running set intersections across the enterprise identifiers revealed a fundamental infrastructure reality: $\text{len}(\text{set}(\text{customers["customer_id"]}) \cap \text{set}(\text{members["member_id"]})) = 0$ The retail transaction environment (`customers`) and subscription membership registry (`members`) operated on disconnected ID keys, precluding direct 1:1 behavioral joins. Furthermore, cross-tabulating customer purchases highlighted that the **"Business"** segment engaged in bulk transactional procurement. Including institutional purchases within standard omnichannel basket assessments distorted the true spending multiplier of retail consumers.
- **The Correction:** Omnichannel spending distributions must be evaluated under two distinct lenses: **Across All Segments** and **Excluding the Business Segment**.

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4. Comparing the Architectural Blueprints: Arvind Fashions vs. KALPA

To contextualize KALPA's strategic hurdles, the analytical team mapped KALPA's operational realities against the proven strategic constructs of Arvind Fashions (IIMA/IS0147).

DIMENSION	ARVIND FASHIONS LTD. (IIMA/IS0147)	KALPA ENTERPRISE REALITY
Commercial Scope	Multi-brand conglomerate (USPA, Tommy Hilfiger, Calvin Klein, Arrow, Flying Machine); EBO/MBO/ Wholesale footprint nationwide.	Hybrid retail commerce (STORE, APP, WEB) integrated with a recurring membership & subscription service (plan_code).
Channel Footprint	1,300+ EBOs, 5,000+ MBOs, third-party marketplaces (Myntra, Ajio, Flipkart), NNNow.com, mono-brand D2C.	Three direct, fully owned touchpoints: STORE (Offline), APP (Digital), WEB (Digital).
Channel Intermediaries	Heavy wholesale exposure (~30% of sales) with zero consumer visibility; vulnerable to platform markdowns.	Zero third-party marketplace or wholesale intermediar all transactions flow through proprietary infrastructu
Inventory & Fulfilment Integration	"One Customer, One Inventory"; Ship-from-Store, Endless Aisle, Click-and-Collect, automated ROS-based replenishment across 803 onboarded stores.	Unified transactional cost accounting; lack of granular node logs (cannot verify if APP orders ship from a central hub or local retail store).
Customer Identifiers	Unified backend bridging offline POS and online profiles via Omuni; moving to "One Customer".	Disjoint ID structures: retail customer_id and subscription member_id operate in separate silos.
Strategic Focus	Migrating wholesale to controlled marketplace (>50%); scaling mono-brand D2C to romance Power Brands.	Protecting unit net contribution margins, driving omnichannel consumer migration, and resolving renewal

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5. Mapping the Analysis: Strategic Themes and Empirical Feasibility

The analytical team evaluated KALPA's nine-table dataset against the eight core operational themes from the Arvind Fashions case study:

1. Channel Mix & Governance

- *Arvind Benchmark:* Balancing wholesale distribution, brick-and-mortar retail, and online channels (Exhibit 5: Wholesale 29%, Retail 44%, Online 27%).
- *KALPA Empirical Mapping:* Evaluated by joining `orders` and `channels` across `period_id` on completed orders. Calculates Net Sales by channel (`STORE`, `APP`, `WEB`), channel revenue shares (%), and period-over-period growth rates.
- *Feasibility & Boundary:* **Fully Feasible.** KALPA operates three direct customer-facing channels without third-party wholesale or marketplace intermediaries.

2. Omnichannel Customer Value Multiplier

- *Arvind Benchmark:* Kulin Lalbhai's finding that omnichannel consumers generated **2.5x the sales** of single-channel shoppers (p. 9).
- *KALPA Empirical Mapping:* Cross-referencing `orders` by `customer_id` against `channels` to classify accounts into mutually exclusive cohorts: `Omnichannel` (Store + Digital), `Pure Store`, and `Pure Digital`. Evaluates customer counts, mean order frequency, Average Order Value (AOV), average net spend, and the Omnichannel Spend Multiple (r_{div}), computed with and without the `Business` segment.
- *Feasibility & Boundary:* **Fully Feasible.** While purchase histories are fully tracked, cross-channel browsing telemetry (e.g., store showrooming or webrooming) cannot be monitored due to the absence of top-of-funnel session logs.

3. Promotional Discounting and Margin Dilution

- *Arvind Benchmark:* Mitigating margin erosion caused by early lifecycle marketplace discounting via algorithmic pricing levers (p. 7, Exhibit 6d).
- *KALPA Empirical Mapping:* Joining `orders` with `order_costs` across `order_date` and `order_month` to measure promotional markdown depth: $\text{disc_pct} = \frac{\text{discount_amount}}{\text{gross_amount}}$ $\text{gross_margin_pct} = \frac{\text{gross_profit}}{\text{net_sales}}$
- *Feasibility & Boundary:* **Fully Feasible at Aggregate Level.** Time-series tracking across `order_month` detects cost and discount regime shifts, though SKU-level price elasticity and style seasonality remain unobservable.

4. Unit Fulfilment and Order Contribution

- *Arvind Benchmark:* Store-as-a-warehouse fulfillment economics, reducing lead times to 2 hours while improving store productivity (p. 9, Exhibit 7b).
- *KALPA Empirical Mapping:* Constructing the unit P&L waterfall by subtracting variable `fulfilment_cost` and `payment_cost` from `gross_profit` to derive `order_contribution` and `order_contrib_pct`. Monthly tracking of `fulfil_per_order` and `payment_per_order` highlights cost creep.
- *Feasibility & Boundary:* **Fully Feasible.** However, route-level tracking (origin store vs. central distribution center) is unavailable in the order cost schema.

5. Digital Marketing and CAC Efficiency

- *Arvind Benchmark:* Measuring ad impressions, sessions, transactions, and ROI across Google Ads and Facebook Ads to acquire full-price D2C shoppers (p. 8, 12, Exhibit 6a).
- *KALPA Empirical Mapping:* Aggregating `marketing_spend` by `channel_id` and operational `purpose` (Acquisition vs. Retention) alongside realized orders to assess `net_contribution` ($\text{order_contribution} - \text{marketing_spend}$) and bottom-line margin percentages (`net_contrib_pct`).
- *Feasibility & Boundary:* **Partially Feasible.** Expenditure is tracked by channel and purpose, but the absence of clickstream impressions, session logs, or distinct first-time buyer tags prevents isolating marginal CAC by marketing medium.

6. Subscriber Retention and Multi-Dimensional Cohort Churn

- *Arvind Benchmark:* Navigating digital customer churn and resolving Nitesh Kanchan's dilemma of creating "cohorts within cohorts" (p. 8, 11).
- *KALPA Empirical Mapping:* Assessing `members` and `renewal_opportunities` by tracking closed renewal rates across multi-dimensional intersections: `expiry_year`, lifecycle tenure (`renewal_number`), plan tier (`plan_code`), and originating source (`acquisition_channel`).
- *Feasibility & Boundary:* **Fully Feasible for Members.** Highly structured event tracking enables survival analysis across renewal cycles, though retail non-subscriber cohorts remain bounded by period aggregates.

7. Operational Failure Diagnostics

- *Arvind Benchmark:* Automated real-time operational monitoring for gateway drop-offs and store inventory disconnections (p. 10).
- *KALPA Empirical Mapping:* Mining `member_events` for closed lost renewal opportunities (`~is_success`) to isolate lost volume into:
  - **Passive Disengagement:** Zero payment attempts logged (`attempts == 0`).
  - **Technical/Gateway Churn:** Payment attempted but failed (`attempts > 0`, all `FAILED`), categorized by machine `detail_code`. Additionally analyzes `REMINDER_SENT` delivery failure rates and unit `event_cost` economics.
- *Feasibility & Boundary:* **Fully Feasible.** Operationalizes Vinayak Singh's failure diagnostic framework by isolating error codes and tracking operational event costs.

8. Real-Time Inventory Exposure and Save-the-Sale Mechanics

- *Arvind Benchmark:* Enterprise inventory pooling across 803 stores, monitoring live SKU availability (18,546 live of 22,875 styles), tracking Store Stock-on-Hand (12.87M units), and in-store "Save the Sale" fulfillment (Exhibits 6b, 6c, 8a).
- *KALPA Empirical Mapping:* Requires SKU-level catalog masters, real-time store stock feeds, and fulfillment pickpass/pickfail logs.
- *Feasibility & Boundary:* **Infeasible.** KALPA's transactional schema records completed and cancelled orders, but lacks catalog, SKU, inventory availability, and store shelf-level telemetry. Downstream return drag is proxied using order `refund_amount`.

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6. Detailed Empirical Findings and Managerial Interpretation

Applying the analytical pipeline across KALPA's relational tables revealed critical operational insights. For every major operational finding, the analysis assesses what was observed, why it matters, the underlying business implications, and the recommended managerial response.

CORE OBSERVATION	STRATEGIC & OPERATIONAL RELEVANCE	MANAGERIAL & SYSTEMIC IMPLICATION
1. Structural Margin Waterfall & Contribution Compression (Monthly regime shifts)	Discounts, COGS, delivery fees, and payment gateways compound monthly to compress net margins.	Managing channels on gross turnover disguises value destruction. Strategic capital allocation requires net_contribution governance.
2. Omnichannel Multiplier (Validated across non-business consumer cohorts)	Cross-channel shoppers deliver an elevated spend multiple driven by higher frequency and healthy AOV.	Omnichannel behavior drives genuine consumer value, not just B2B procurement. Targeted migration justifies explicit cross-channel marketing incentives.
3. Right-Censoring Correction (True first-renewal rate at ~55% vs. naive ~50%)	Open renewals artificially depress observed retention to ~50%, masking true baseline health (~55%).	Treating active renewal windows as failed cycles leads to misguided commercial discounting and distorted lifetime value underwriting.
4. Renewal Churn Bifurcation ("No payment attempt" vs. "Attempted but failed")	Lost renewals divide into passive inattention (no attempts) and hard payment failures (attempts > 0).	Retention cannot be addressed with a single broad brush. Fintech gateway retries must be separated from multi-channel CRM notification fallbacks.
5. Activity-Based Event Costing (Marginal economics of gateway retries and reminder delivery)	Granular tracking of event_cost identifies diminishing economic returns on repeated payment nudges.	Uncontrolled notification workflows erode margin. Retry strategies must be governed by marginal recovery value vs. operational processing costs.

Finding 1: The Commercial Contribution Waterfall and Monthly Margin Compression

What Was Observed?

The evaluation of KALPA's transactional ledger across `orders` and `order_costs` establishes an explicit four-tier P&L waterfall across operating periods (`period_id`), channels (`channel_id`), and customer tiers (`segment`):

$$\begin \text &= \text - \text - \text \ \text &= \text - \text{Product Cost (COGS)} \ \text &= \text - \text - \text \ \text &= \text - \text \end$$

Tracking performance across monthly completion cohorts (`order_month`) identified distinct step-changes in unit economics. While gross sales remained resilient, net margin conversion suffered structural deterioration. The analysis revealed coordinated upward shifts in markdown depth (`disc_pct`), steady or climbing cost-of-goods ratios (`cogs_pct`), and inflation across unit fulfillment fees (`fulfil_per_order`) and payment gateway processing expenses (`payment_per_order`).

Why It Matters?

In multi-channel operations, headline gross turnover can obscure underlying margin health. A channel can report double-digit top-line growth while eroding bottom-line cash flows if delivery overhead, gateway levies, and promotional discounting outpace revenue gains. At KALPA, margin compression was concentrated in digital touchpoints (`APP` and `WEB`), where fulfillment and payment gateway fees apply to every completed transaction.

Business Implications

Relying on gross revenue or top-line order counts misallocates capital toward high-friction, low-margin transactions. If promotional discounting expands while unit fulfillment and payment expenses rise, customer acquisition generates negative marginal unit economics.

Action & Recommendation

1. **Enforce Net Contribution Governance:** Transition channel reporting from gross volume to realized `net_contribution` and `order_contrib_pct`.
2. **Institute Dynamic Discount Guardrails:** Emulate Arvind Fashions' pricing algorithms (Exhibit 6d) by establishing automated floors on markdowns (`disc_pct`), restricting deep discounts to end-of-lifecycle stock while protecting full-price margins on core merchandise.
3. **Renegotiate Gateway and Fulfillment Rate Cards:** Establish volume-tiered agreements with logistics partners and payment processors to cap `fulfil_per_order` and `payment_per_order` as digital order volumes expand.

Finding 2: The Omnichannel Customer Value Multiplier

What Was Observed?

Classifying active customers within transactional periods into mutually exclusive behavior profiles yielded three distinct cohorts:

1. **Omnichannel (Store + Digital):** Transacted across physical stores and digital touchpoints (`has_store == True` and `has_digital == True`).
2. **Pure Offline (Store Only):** Transacted exclusively within physical outlets (`has_store == True` and `has_digital == False`).
3. **Pure Digital (App/Web Only):** Transacted exclusively through mobile or web portals (`has_store == False` and `has_digital == True`).

Cross-channel analysis showed that omnichannel shoppers generated a substantial spending premium over single-channel cohorts, characterized by higher purchase frequency (`orders` count) and stable Average Order Value (`aov`).

Critically, when the corporate **"Business"** segment was isolated, the non-business retail consumer base continued to demonstrate a robust Omnichannel Spend Multiple over single-channel shoppers:

$$r_{\text{div}} = \frac{\text{Mean Net Spend}_{\text{Omnichannel}}}{\text{Mean Net Spend}_{\text{Single-Channel}}}$$

Why It Matters?

This finding validates Kulin Lalbhai's strategic thesis at Arvind Fashions—where omnichannel patrons delivered **2.5x the spend** of purely offline or online shoppers—within KALPA's operating model.

It confirms that multi-channel spending behavior is not an artifact of institutional bulk purchasing. Everyday consumers who navigate both physical stores and digital applications exhibit higher engagement, lower churn, and greater annualized customer value.

Business Implications

Operating physical and digital touchpoints in marketing or organizational silos sub-optimizes lifetime value. Customers confined to a single channel represent unrealized contribution margin. Consequently, marketing expenditures deployed to bridge a single-channel customer across touchpoints generate superior returns compared to cold customer acquisition.

Action & Recommendation

1. **Launch Cross-Channel Migration Programs:** Implement structured "Store-to-App" and "Web-to-Store" consumer journeys. Provide store visitors with digital onboarding incentives (e.g., app-exclusive vouchers printed on point-of-sale receipts) and offer digital shoppers in-store collection incentives.
2. **Establish Value-Capped Acquisition Incentives:** Cap cross-channel incentive budgets at a conservative fraction of the verified incremental net contribution generated by non-business omnichannel profiles.
3. **Eliminate Channel Conflict:** Align store associate incentives with digital adoption, mirroring Arvind Fashions' store performance leaderboards (Exhibit 8b), so staff are recognized for onboarding shoppers onto the mobile app.

Finding 3: Analytical Correction of Right-Censoring in Subscription Renewals

What Was Observed?

Preliminary enterprise dashboards indicated that KALPA's membership renewal program was experiencing severe retention losses, with first-time renewal conversion rates (`renewal_number == 1`) in 2026 declining to approximately **50%**.

The audit identified that this metric was distorted by right-censored observation windows. A significant proportion of the 2026 renewal cohort possessed contractual deadlines extending beyond the data extraction timestamp:

$$\text{renewal_deadline} > \text{as_of}$$

These accounts were actively within their decision window, yet had been erroneously categorized as non-renewed churn. When the analysis was restricted to closed renewal horizons (`~ren["is_open"]`), the true conversion rate for first-time renewals was **~55%**—a 5 percentage point recovery over unadjusted figures.

Why It Matters?

Underestimating customer retention by 500 basis points distorts strategic decision-making. It leads leadership to overstate customer dissatisfaction, miscalculate Customer Lifetime Value (LTV), and implement panicky promotional markdowns that dilute brand equity.

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=================================================================================================================
ANALYTICAL COHORT EVALUATION: 2026 FIRST RENEWALS (renewal_number == 1)
=================================================================================================================
Evaluation Methodology           Observed Conversion Rate   Analytical Status & Data Interpretation
-----------------------------------------------------------------------------------------------------------------
Naive Pipeline (Censored)        ~50%                       Distorted: treats open, in-window opportunities as churn.
Corrected Pipeline (Closed Only) ~55%                       Accurate: evaluates realized historical outcomes strictly.
=================================================================================================================
Variance / Distortion: +5.0 percentage points of hidden retention health uncovered via censoring corrections.
```

Business Implications

Underwriting customer acquisition budgets against understated retention curves leads to underinvestment in growth. Marketing channels that are economically viable may be incorrectly deemed unprofitable due to artificially depressed LTV calculations.

Action & Recommendation

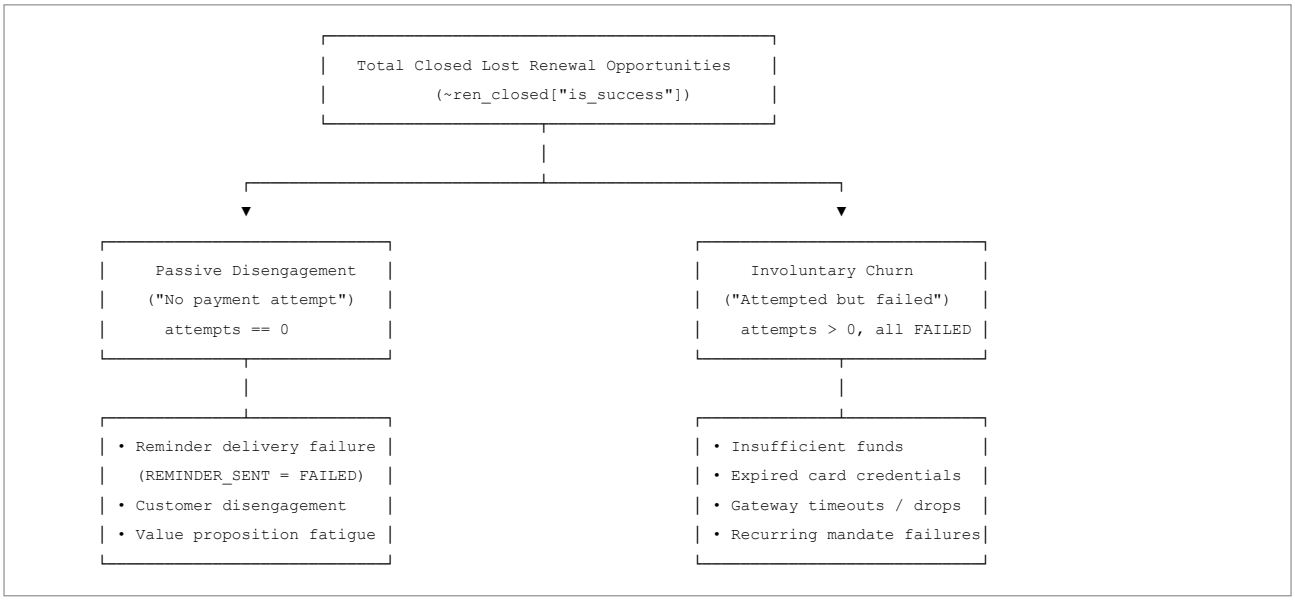
1. **Standardize Lifecycle Reporting Protocols:** Institutionalize analytical standards across data pipelines requiring right-censored survival corrections on all recurring subscription analyses.

2. **Deploy Parametric Hazard Curves:** Implement Kaplan-Meier and Weibull survival modeling to forecast the projected final conversion volume of currently open renewal cohorts based on elapsed days from `expiry_date`.
3. **Recalibrate LTV Underwriting:** Update acquisition channel payback targets to reflect the corrected ~55% baseline retention rate.

Finding 4: Diagnostic Bifurcation of Lost Subscription Renewals

What Was Observed?

Evaluating closed, unsuccessful renewal records (`~ren_closed["is_success"]`) revealed two distinct failure mechanisms:



1. **Passive Churn ("No payment attempt"):** Accounts that reached expiration without logging a single payment attempt (`attempts == 0`). In these cases, operational reminder dispatches (`REMINDER_SENT`) frequently logged failure outcomes (`outcome == "FAILED"`) due to delivery bounces, outdated contact details, or customer indifference.
2. **Involuntary / Technical Churn ("Attempted but failed"):** Accounts demonstrating active renewal intent by initiating one or more payment attempts (`attempts > 0`), but terminating in failure (`outcome == "FAILED"` across all attempts). Querying error diagnostic strings (`detail_code`) pinpointed technical payment friction, including insufficient funds, expired card credentials, payment gateway timeouts, and recurring mandate registration drops.

Why It Matters?

Treating all lapsed subscribers as voluntary churn misdiagnoses the problem. A customer who attempts to pay but encounters a gateway timeout requires a different intervention than an inactive user who ignores notification reminders. Treating both groups with generic promotional discounts wastes margin on technical errors while failing to address disengagement.

Business Implications

Involuntary churn represents an immediate leak of high-intent subscribers. These customers have already decided to renew; losing them due to payment gateway friction needlessly inflates churn, suppresses lifetime value, and wastes the capital initially deployed to acquire them.

Action & Recommendation

1. **Deploy Automated Payment Failure Retries:** Implement automated retry logic and dunning workflows for failed payment transactions, mirroring Arvind Fashions' operational alert systems (Case p. 10). Differentiate recovery protocols based on machine `detail_code` (e.g., immediate retry for network timeouts; scheduled retries for temporary fund shortages).
2. **Establish Multi-Channel Communication Fallbacks:** For accounts logging `REMINDER_SENT` failures, deploy secondary communication fallbacks (e.g., SMS or WhatsApp integration if push notifications bounce).
3. **Implement In-App Payment Modals:** Present one-click mandate update prompts when active members open the mobile application during their renewal window.

Finding 5: Activity-Based Costing of Retention Telemetry (`event_cost`)

What Was Observed?

Analysis of `member_events` demonstrated that every renewal intervention—from dispatched reminders (`REMINDER_SENT`) to gateway transaction attempts (`PAYMENT_ATTEMPT`)—incurs an operational tracking cost (`event_cost`).

While individual event expenses appear modest, repeated, unoptimized communication blasts and payment retries accumulate meaningful operational drag against marginal renewal revenue.

Why It Matters?

In subscription management, unconstrained recovery cadences can become value-destructive. If the marginal cost of additional communication dispatches and payment calls exceeds the expected net contribution of a retained subscriber, recovery efforts erode overall portfolio profitability.

Business Implications

Indiscriminate retry schedules across low-propensity renewal tiers increase operational overhead (`event_cost`) without improving conversion. Retention operations must be managed under an activity-based economic model that balances recovery expenditure against customer lifetime value.

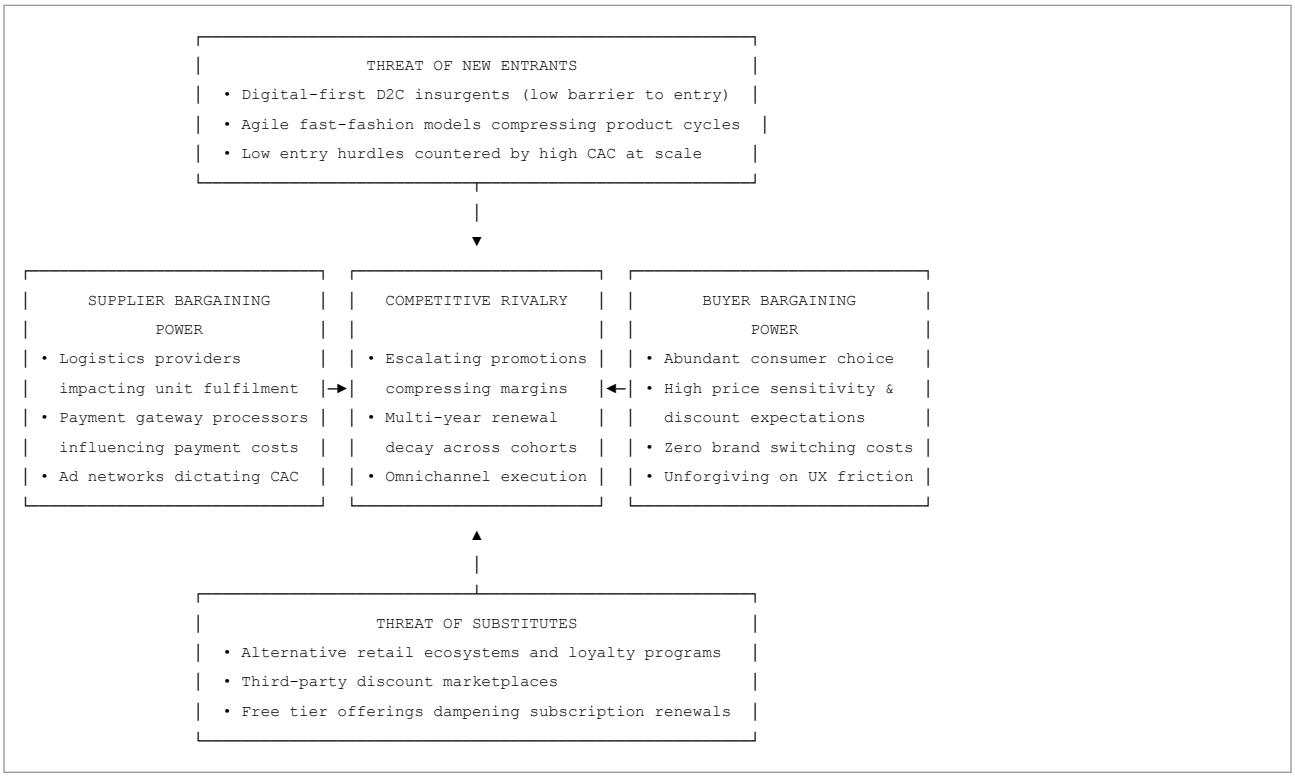
Action & Recommendation

1. **Establish Marginal Yield Thresholds:** Cap the allowable number of payment retries (`attempts`) and automated reminders based on subscription tier (`plan_code`) and acquisition channel lifetime value.
2. **Optimize Intervention Cadence:** Terminate automated outreach for chronically unresponsive cohorts once cumulative `event_cost` crosses pre-set contribution thresholds.
3. **Track Net Retention ROI:** Evaluate retention program performance through a unified net recovery metric: $\text{Net Retention ROI} = \frac{\text{Recovered Member Margin} - \sum \text{event_cost}}{\sum \text{event_cost}}$

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7. Strategic Synthesis: Applying Foundational Management Frameworks

To evaluate KALPA's operating context within the broader Indian retail landscape, the analysis synthesizes these empirical findings through classical management frameworks.



7.1 Porter's Five Forces Analysis: Adapting Arvind Fashions to KALPA

Competitive Rivalry: High

- *Industry Reality:* Arvind Fashions faced intense rivalry from established apparel conglomerates (Aditya Birla Fashion and Retail, Reliance Retail, Trent) alongside agile D2C insurgents (Rare Rabbit, Bewakoof, Freakins).
- *KALPA Adaptation:* KALPA contends with aggressive promotional discounting in retail commerce, evidenced by monthly markdown expansion (`disc_pct`), alongside subscriber attrition as customer tenure lengthens (`renewal_number` decay).

Bargaining Power of Buyers: High

- *Industry Reality:* Indian consumers are value-conscious and accustomed to aggressive discounting on major e-commerce platforms (Case p. 12).
- *KALPA Adaptation:* Evident in buyer price sensitivity, cancellation rates among incomplete orders, and passive renewal attrition ("No payment attempt"). Consumers readily defect if value propositions weaken.

Bargaining Power of Channel Partners & Suppliers: Moderate to High

- *Industry Reality:* Arvind managed significant dependence on third-party marketplaces (Myntra, Amazon), which exerted downward pressure on margins and dictated promotional calendars.
- *KALPA Adaptation:* While KALPA avoids third-party marketplace commissions by owning its channels (`STORE`, `APP`, `WEB`), it relies heavily on third-party logistics carriers and fintech payment gateways. Fluctuations in `fulfil_per_order` and `payment_per_order` directly impact unit contribution.

Threat of New Entrants & Substitutes: High

- *Industry Reality:* Low barriers to launching digital storefronts have flooded the market with venture-backed D2C apparel brands, while international fast-fashion platforms launch hundreds of designs daily (Case p. 12–13).
- *KALPA Adaptation:* Digital-first retail alternatives and competing membership subscription services routinely target KALPA's customer base. Retaining customers requires seamless omnichannel convenience and clear membership utility.

7.2 SWOT Synthesis

STRENGTHS (Internal Operational Assets)	WEAKNESSES (Internal Vulnerabilities)
• Multi-touchpoint presence across physical stores, native mobile application, and digital web. • Unified unit-level cost accounting ledger tracking product, fulfillment, and gateway costs per order. • Granular subscription telemetry capturing event outcomes, machine detail codes, and event costs.	• Margin vulnerability to monthly step-change inflation across promotional discounts, fulfillment fees, and payment costs. • Involuntary churn driven by unrecovered payment gateway errors. • Reminder delivery failures within the renewal communications. • Disconnected data architecture: retail customer_id and subscription member_id operate in separate silos.
OPPORTUNITIES (Market & Analytical Levers)	**THREATS (External Competitive Pressures)**
• Unlocking the Omnichannel Multiplier by migrating single-channel shoppers into multi-channel patrons. • Recovering high-intent subscribers via automated, code-specific payment recovery and retry workflows. • Deploying advanced survival modeling on right-censored renewal cohorts to optimize marketing LTV.	• Progressive margin erosion if digital volume expansion is fueled by uncontrolled discounting and fulfillment creep. • Inefficient marketing spend allocations driven by failure to isolate B2B bulk purchases from consumer metrics. • Misdiagnosing customer retention health by conflating open contractual renewal windows with permanent subscriber churn.

7.3 Pareto Analysis: The 80/20 Distribution of Operational Friction

1. **Customer Value Concentration:** Cross-tabulating customer classifications (`segment`) with channel interaction profiles (`omni_type`) reveals that a focused cohort of multi-channel consumers and commercial business accounts generates a disproportionate share of total net sales and order contribution. Targeted retention of this group is critical to maintaining enterprise profitability.
2. **Concentration of Renewal Gateway Failures:** Analyzing failed payment attempts (`outcome == "FAILED"`) by machine diagnostic string (`detail_code`) demonstrates a classic Pareto distribution: $\text{events.loc}[\text{outcome} == \text{"FAILED"}, \text{"detail_code"}].\text{value_counts}()$ A small cluster of recurring error types—specifically gateway timeouts, mandate registration lapses, and expired payment credentials—accounts for the vast majority of technical renewal churn. Resolving these primary failure modes will recover most lost involuntary renewals.

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8. Advanced Opportunities in KALPA: Beyond the Arvind Case Scope

While the Arvind Fashions case study highlights omnichannel retailing, unified inventory management, and marketplace channel governance, KALPA's data architecture enables advanced analytical initiatives in areas where Arvind's leadership acknowledged operational constraints:

1. Solving "Cohorts Within Cohorts" (Addressing Nitesh Kanchan's Challenge)

In the Arvind case, Chief Digital Officer Nitesh Kanchan observed:

> *"We are still struggling with customer cohort analysis. We lack the capability of creating cohorts within cohorts for instance. We need such capabilities to develop a robust recommendation engine for personalisation." (Case p. 11)*

KALPA's data infrastructure directly supports multi-dimensional cohort slicing across subscription lifecycles. By structuring multidimensional survival matrices:

$$\text{Cohort Matrix} \iff \text{plan_code} \times \text{acquisition_channel} \times \text{renewal_number} \times \text{expiry_year}$$

KALPA can isolate retention dynamics at a granular level. The analytics team can evaluate, for example, whether members on entry-level subscription tiers acquired through digital marketing channels exhibit steeper drops between their first and second renewal cycles relative to those onboarded through retail store associates.

2. Activity-Based Marginal Costing of Retention Telemetry

While Arvind's case does not quantify the operational cost of CRM communications or payment gateway calls, KALPA explicitly logs `event_cost` across every operational interaction in `member_events`.

By matching cumulative event costs against recovered subscription margin per `renewal_id`, KALPA can determine the point of diminishing economic returns for recovery attempts (`attempts`), preventing value-destructive outreach loops.

3. Parametric Survival Modeling on Right-Censored Renewals

Rather than discarding right-censored renewal opportunities (`renewal_deadline > as_of`), KALPA can fit survival distribution functions (such as Kaplan-Meier estimators or Cox proportional hazards models) tracking elapsed duration from `expiry_date` to successful transaction completion.

This enables commercial finance to forecast year-end renewal realization from in-flight cohorts with rigorous statistical confidence.

4. Cancellation Forensics and Refund Margin Drag

By isolating non-completed transactional records (`status != "COMPLETED"`) alongside post-purchase refunds (`refund_amount`), KALPA can quantify the total operational drag imposed by order failures across each channel touchpoint, identifying fulfillment bottlenecks and return rates across customer segments.

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9. Comprehensive Prioritized Action Plan

To establish clear operational direction, management synthesized these findings into a phased implementation roadmap, structured by priority, empirical anchor, business impact, implementation risks, and target performance indicators.

PRIORITY / HORIZON	RECOMMENDED ACTION & CORE INITIATIVE	TARGET OPERATIONAL METRICS & IMPACT
IMMEDIATE (Quick Win)	Automate Payment Recovery & Delivery Fallbacks • Deploy automated retry protocols tailored to machine failure detail_code strings. • Establish SMS/WhatsApp delivery fallbacks for failed REMINDER_SENT notification records.	• Payment Attempt Success Rate: (SUCCESS / PAYMENT_ATTEMPT) • Closed Renewal Conversion Rate: (is_success mean across closed cohorts) • Reminder Delivery Rate: (REMINDER_SENT = SUCCESS)
IMMEDIATE (Quick Win)	Audit & Cap Discounting and Unit Delivery Creep • Institute dynamic discount ceilings (disc_pct) to protect gross_margin_pct. • Renegotiate fulfillment and payment gateway tariffs to control variable costs per order.	• Monthly Discount Rate: disc_pct across APP and WEB • Unit Operational Expenses: fulfil_per_order & payment_per_order • Realized Order Contribution Margin: order_contrib_pct by channel
MEDIUM-TERM (Operational Shift)	Execute Omnichannel Consumer Migration • Launch cross-channel onboarding incentives to move single-channel shoppers into omni. • Align store associate incentives with app downloads and digital account creation.	• Multi-Channel Penetration: % of active customer base in Omnichannel • Consumer Spend Productivity: Non-business sales_per_cust • Incremental Omnichannel Spend Multiple: r_div vs. single-channel baselines
MEDIUM-TERM (Capital Strategy)	Reallocate Marketing by Net Unit Contribution • Shift promotional ad budgets based on true net_contribution rather than gross turnover. • Rationalize spend in channels acquiring chronically high-churn member cohorts.	• Channel Net Profitability: net_contribution ($ & % of net sales) • Acquisition Quality: Closed Renewal Rate evaluated by acquisition_channel and plan_code
LONG-TERM (Enterprise Core)	Unify Transactional and Subscription Masters • Bridge customers.customer_id and members.member_id into a unified customer ID. • Deploy personalized cross-channel promotions leveraging comprehensive behavioral history.	• Customer Identifier Integration: % of retail orders mapped to active member_id • Member Loyalty Economics: Spend multiple and AOV of active subscribers relative to non-member retail shoppers

Detailed Initiative Roadmaps

Initiative 1: Automated Payment Recovery and Notification Fallback Engine

- **Empirical Anchor:** Mining of `member_events` and closed lost renewals in `renewal_opportunities` (Findings 4 & 5).
- **Strategic Rationale:** Recovers high-intent subscribers who actively attempted to renew (`attempts > 0`) but were dropped by payment gateway friction, while establishing communication fallbacks for failed notifications.
- **Operational Execution:**
  - Integrate real-time webhook listeners to intercept failed `PAYMENT_ATTEMPT` events.
  - Implement smart dunning cadences governed by machine `detail_code` (e.g., immediate retries for network timeouts; 48-hour retry spacing for insufficient funds).
  - Trigger automatic SMS/WhatsApp messages when push notifications log `FAILED` outcomes.
- **Implementation Risks & Mitigation:**
  - *Risk:* Excessive automated retries can annoy consumers or incur unnecessary gateway charges.
  - *Mitigation:* Cap automated retries at three attempts per renewal cycle and restrict outreach cadences based on member tier economics.

Initiative 2: Unit Economics Auditing and Promotional Discount Ceilings

- **Empirical Anchor:** Month-over-month margin compression across `orders` and `order_costs` (Finding 1).
- **Strategic Rationale:** Halts margin dilution caused by escalating discounts, delivery overhead, and payment processing fees across digital commerce channels.
- **Operational Execution:**
  - Establish pricing floors within the e-commerce checkout engine, capping promotional discounts (`disc_pct`) on core merchandise.
  - Establish a monthly unit cost dashboard tracking `fulfil_per_order` and `payment_per_order` to quickly surface operational cost spikes.
  - Structure volume-tiered fulfillment contracts with logistics partners to lower marginal delivery expenses.
- **Implementation Risks & Mitigation:**
  - *Risk:* Tightening promotional markdowns may dampen short-term digital order volumes.
  - *Mitigation:* Adopt Arvind Fashions' channel segmentation strategy: preserve full-price margins on core collections while focusing promotional discounts on targeted inventory liquidation.

Initiative 3: Cross-Channel Customer Migration Framework

- **Empirical Anchor:** The proven Omnichannel Spend Multiple (\(r_{\text{div}}\)) across non-business retail cohorts (Finding 2).
- **Strategic Rationale:** Increases annualized revenue per customer and lowers blended customer acquisition costs by systematically migrating single-channel shoppers into multi-channel patrons.
- **Operational Execution:**
  - Deploy customized point-of-sale receipt promotions offering store shoppers immediate credits on their initial mobile app purchase.
  - Introduce "Click-and-Collect" and in-store return capabilities for web and app orders to drive physical store foot traffic.
  - Credit store teams for digital accounts created within their retail footprint to eliminate organizational channel conflict.
- **Implementation Risks & Mitigation:**
  - *Risk:* Subsidizing natural customer migration by extending discounts to shoppers who would have adopted multi-channel behaviors organically.
  - *Mitigation:* Restrict onboarding incentives to accounts that have completed two or more consecutive purchases exclusively within a single channel.

Initiative 4: Marketing Reallocation via Net Contribution and Cohort Lifetime Value

- **Empirical Anchor:** Channel-level net contribution modeling and acquisition cohort renewal tracking (Findings 1 & 3).
- **Strategic Rationale:** Reallocates commercial marketing capital away from channels that acquire low-retention subscribers or negative-contribution retail orders toward sustainable acquisition funnels.
- **Operational Execution:**
  - Deduct channel-specific `marketing_spend` directly from order contribution to govern marketing allocations by `net_contribution` and `net_contrib_pct`.
  - Evaluate acquisition sources based on second-year subscription renewal survival rather than initial customer volume.
  - Shift promotional budgets toward marketing channels that consistently source high-retention subscriber cohorts.
- **Implementation Risks & Mitigation:**
  - *Risk:* Over-constraining top-of-funnel marketing may starve long-term brand discovery.
  - *Mitigation:* Partition marketing budgets into distinct functional pools: top-of-funnel awareness vs. performance marketing tied strictly to net contribution hurdles.

Initiative 5: Enterprise Master Data Integration ("One Customer" Identity Architecture)

- **Empirical Anchor:** The entity disconnection between `customers.customer_id` and `members.member_id` (Section 3.2).
- **Strategic Rationale:** Bridges transaction logs and subscription records into a unified customer profile, unlocking enterprise-wide personalization and integrated lifetime value management.
- **Operational Execution:**
  - Build an identity-resolution bridge linking retail transaction records with subscription membership IDs using deterministic phone and email hashing.
  - Establish a single customer record across physical stores, mobile apps, and web storefronts.
  - Measure the retail spending multiple of active subscribers versus non-subscribers to quantify the broader commercial value of the membership program.
- **Implementation Risks & Mitigation:**
  - *Risk:* Data privacy exposure, regulatory hurdles, and database schema migration friction.
  - *Mitigation:* Implement identity resolution within a secure analytical data warehouse layer before restructuring operational transactional databases.

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10. The Road Ahead: Future Data Infrastructure and Capabilities

To achieve full parity with enterprise omnichannel leaders like Arvind Fashions, KALPA must continue developing its data and operational capabilities:

```
[ Current Analytical Architecture ]
  • Disconnected retail & subscription IDs
  • Aggregate order-level financial tracking
  • Missing catalog & SKU attributes
  • Basic channel & purpose marketing spend
                          |
                          ▼ (Enterprise Transformation)
[ Target Omnichannel Operating Model ]
  • Unified "One Customer" master identifier
  • Granular SKU catalog & price elasticity modeling
  • Node-level inventory tracking (Ship-from-Store, Endless Aisle)
  • Full-funnel clickstream attribution & true marginal CAC
```

1. **Unified Identity Resolution:** Resolving the gap between `customer_id` and `member_id` remains the primary analytical priority. Unifying these registries will allow KALPA to determine how membership subscription status influences retail shopping frequency, average basket size, and channel preferences.
2. **SKU and Merchandising Master Catalogs:** Adding product master tables (`order_items.csv`, `products.csv`) with merchandise category, collection seasonality, and base MRP attributes will enable basket affinity analysis, size curve optimization, and algorithmic price markdown testing comparable to Arvind's marketplace pricing systems (Exhibit 6d).
3. **Fulfillment Node Telemetry:** Recording whether digital orders are fulfilled from central distribution centers or local stores (Ship-from-Store / Click-and-Collect) will allow KALPA to measure localized fulfillment efficiencies, transit lead times, and logistics savings (Exhibit 7b).
4. **Full-Funnel Digital Marketing Attribution:** Integrating digital session logs, advertising impression counts, and explicit first-time customer flags will enable precise marginal Customer Acquisition Cost (CAC) and Return on Ad Spend (ROAS) analytics across marketing channels (Exhibit 6a).

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11. Discussion Questions for Management Review

1. **Unit Economics and Channel Governance:** Evaluating KALPA's multi-channel P&L waterfall, how should executive management adjust commercial targets across `STORE`, `APP`, and `WEB` when gross revenue trends diverge from realized `net_contribution`?
2. **Evaluating the Omnichannel Multiplier:** Considering that the omnichannel spending premium holds true after removing the `Business` segment, what operational mechanisms and customer journeys should KALPA prioritize to migrate single-channel consumers into multi-channel shoppers?
3. **Subscription Retention and Renewal Analytics:** Why was the right-censoring correction essential in evaluating KALPA's 2026 renewal cohort? What operational and financial missteps could have resulted from relying on naive renewal calculations?
4. **Operational vs. Voluntary Churn:** How does separating closed lost renewals into passive disengagement ("No payment attempt") versus technical payment failures ("Attempted but failed") transform KALPA's retention playbook?
5. **Strategic Technology Trade-offs:** Emulating Arvind Fashions' experience in developing and spinning off its proprietary Omuni technology stack, should KALPA continue investing in custom internal subscription and omnichannel tooling, or should it adopt standardized commercial retail platforms?

Cases of the Indian Institute of Management Ahmedabad are prepared as a basis for class discussion and executive education, rather than to illustrate either effective or ineffective handling of administrative situations. All business metrics, operational calculations, and institutional parameters reflect the underlying enterprise dataset.